

$O(n)$ -size, $O(\log n)$ -depth circuits are computable by
depth-3 circuits of size $2^{O(n/\log \log n)}$
Valiant's depth reduction

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1 Introduction

In these study notes, we examine the main result of the 1977 paper titled “Graph-theoretic arguments in low-level complexity”, where Valiant showed that a linear-size, logarithmic-depth Boolean circuit can be converted into a subexponential-size depth-3 circuit [8]. More specifically, Valiant showed the following:

Theorem 1.1 (Valiant's depth reduction). *Suppose that $f : \{0, 1\}^n \rightarrow \{0, 1\}$ can be computed by a Boolean circuit of linear size and logarithmic depth. Then, for any $\epsilon > 0$, f can also be computed by a circuit with the following properties:*

1. *The circuit is an OR of $2^{O(n/\log \log n)}$ CNFs, each of which has size $2^{O(n^\epsilon)}$.*
2. *For every input, at most one of the CNFs outputs true.*

Remark 1.1. In other words, f can be computed by a circuit of type Σ_3 , size $2^{O(n/\log \log n)}$, and depth 3 in negation-normal form. Per the convention, a circuit can have AND, OR, and NOT gates. In constant-depth circuits, AND and OR gates have unbounded fan-ins, and NOT gates do not count toward depth. Otherwise, AND and OR gates have 2 fan-ins, and NOT gates do count toward depth.

The main motivation behind this work was the following unresolved problem in the field of circuit complexity:

Find an explicit Boolean function that cannot be computed by a linear-size circuit.

When the explicit requirement is relaxed, the question can be resolved in the positive using a trivial counting argument. Otherwise, the best known explicit circuit lower bound for the basis U_2 (AND/OR gates with free NOT gates) is $5n - o(n)$ due to Iwama and Morizumi [6]. Despite decades of research, the task of proving superlinear circuit lower bounds remains elusive.

Instead, Valiant frames the following related problem as another fundamental circuit-lower bound barrier:

Find an explicit Boolean function that cannot be computed by a linear-size, **logarithmic-depth** circuit.

Furthermore, Valiant reduces the related problem to another that seems slightly easier: It suffices to show an exponential circuit lower bound for depth-3 circuits. Unfortunately, even with this severe logarithmic-depth restriction, Valiant's promised lower-bound application remains unrealized to this day. Indeed, the best known explicit circuit lower bound for unbounded-fan-in depth-3 circuits is $2^{\Omega(\sqrt{n})}$ due to Håstad, Jukna, and Pudlák [5].

Nonetheless, Valiant's depth reduction has had a profound impact in the field of computational complexity. In these study notes, we give a proof of Theorem 1.1.

2 Graph-theoretical proof ingredients

Theorem 1.1 is largely a simple application of an elegant graph-theoretical result, which we introduce in this section. A circuit can be represented by a directed acyclic graph (DAG), so, in our analysis, we only consider DAGs, whose depth is defined as follows:

Definition 2.1. The **depth** of a DAG is the length of its longest path.

We also define graph labeling, as follows:

Definition 2.2. A **labeling** of a DAG with vertices V and edges E is a mapping $\ell : V \rightarrow \mathbb{N}$. A labeling ℓ is **legal** if $\ell(u) < \ell(v)$ for all edges $(u, v) \in E$.

The canonical labeling scheme plays an important role in our analysis:

Definition 2.3. The **canonical** labeling of a DAG with vertices V is a mapping $\ell : V \rightarrow \mathbb{N}$ where $\ell(v)$ is the length of the longest path ending at v .

In plain terms, the canonical labeling assigns a vertex its depth as its label. This is the labeling introduced in the proof of Theorem 5.1 in Valiant's paper [8]. While Valiant did not deeply analyze graph labeling, we do so both as a warm-up and to facilitate a graph-theoretic analysis.

First, we show the following:

Proposition 2.4. The canonical labeling is legal.

Proof. Consider a DAG with vertices V and edges E . By Definition 2.3, we have that, for all $v \in V$ such that $\exists u (u, v) \in E$,

$$\ell(v) = \max_{u \in V: (u, v) \in E} \ell(u) + 1 > \max_{u \in V: (u, v) \in E} \ell(u).$$

This implies $\ell(u) < \ell(v)$ for all $(u, v) \in E$, satisfying Definition 2.2. □

Next, we show the following relations between a DAG's depth and labeling:

Proposition 2.5. Let G be a DAG. G has a legal labeling using at most d distinct labels if and only if its depth is less than d .

Proof. We begin by showing the **forward** direction. Suppose for the sake of contradiction that G has a legal labeling using at most d distinct labels and its depth is at least d . Let ℓ be one such legal labeling. There exists a path in G consisting of d edges and hence $d + 1$ nodes. Let $v_1, v_2, \dots, v_{d+1}$ be the nodes encountered by traversing this path. For all $i \in [d]$, we have that $\ell(v_i) < \ell(v_{i+1})$ since the pair of nodes is connected by an edge. Thus, $\ell(v_1), \ell(v_2), \dots, \ell(v_{d+1})$ are strictly and monotonically increasing, so the number of distinct labels in ℓ is at least $d + 1$, contradiction.

We continue by showing the **backward** direction. Suppose that G has depth less than d . Consider the canonical labeling, which we showed is legal in Proposition 2.4, and denote it again by ℓ . The length of the longest path ending at any node is less than d . By definition 2.3, we have that all labels in ℓ are less than d . Since lengths are natural numbers, ℓ uses at most d distinct labels. $\square$

Consider representing the labels in binary. Then, we have the following proposition:

Proposition 2.6. *Let $G = (V, E)$ be a DAG, and consider the canonical labeling ℓ on G in binary. For all $j \in \mathbb{N}$, let E_j be the set of edges (u, v) such that the most significant bit in which $\ell(u)$ and $\ell(v)$ differ is the j^{th} bit. For any $j \in \mathbb{N}$, if the edges in E_j are deleted from G , deleting the j -th bit from ℓ yields a legal labeling of the new graph.*

Proof. Denote the new graph by $G' = (V, E')$ where $E' = E \setminus E_j$, and let ℓ' be the new labeling. By Definition 2.2, it suffices to show that $\ell'(u) < \ell'(v)$ for all $(u, v) \in E'$. Suppose for the sake of contradiction that $\ell'(u) \geq \ell'(v)$ for some $(u, v) \in E'$. By Definition 2.2, we have that $\ell(u) < \ell(v)$. Clearly, the deletion of the j -th bit flips the inequality of the labels. This is only possible if the j -th bit is the most significant bit at which $\ell(u)$ and $\ell(v)$ differ, so we have that $(u, v) \in E_j$. Then, $E' = E \setminus E_j \not\ni (u, v)$, contradiction. $\square$

We are now ready to show our main graph-theoretical result, concerning depth reduction, which corresponds to Corollary 5.2 in Valiant's paper [8].

Lemma 1. *Let G be a DAG with m edges and depth less than d . For the sake of simplicity, assume that d is a power of 2, i.e., $d = 2^k$ for some k . Then, for any integer $1 \leq r \leq k$, the depth of G can be reduced to less than $d/2^r$ by removing at most $(r/k) \cdot m$ edges.*

Proof. We prove this by strong induction on k . Let k be a natural number. Suppose that for all natural numbers $k' < k$, for all DAGs G' with m' edges and depth less than $d' = 2^{k'}$, and for any integer $1 \leq r' \leq k'$, the depth of G' can be reduced to less than $d'/2^{r'}$ by removing at most $(r'/k') \cdot m'$ edges. ($\star$)

Let V and E be the set of vertices and m edges in G . For all $j \in \mathbb{N}$, define E_j as in Proposition 2.6. Define j^* such that E_{j^*} is the smallest set among $E_0, E_1, \dots, E_{k-1}$ (since $1 \leq r \leq k$, there is a non-zero number of sets). Consider the procedure that removes all edges in E_{j^*} from E , resulting in a new graph $G' = (V, E \setminus E_{j^*})$. Let $t = |E_{j^*}|$. We selected the smallest set, so the number of removed edges is at most the average size of the sets, i.e., $t \leq m/k$.

Let ℓ be the canonical labeling of G . By Proposition 2.4, ℓ is a legal labeling on G . By Proposition 2.5, we have that ℓ uses at most d distinct labels because the depth of G is less than d . By Proposition 2.6, we have that deleting the j^* -th bit from all labels in ℓ results in a legal labeling on G' , and let us denote this by ℓ' . Since ℓ' uses 1 fewer bit than ℓ , we have that ℓ' uses at most $d/2$ distinct labels. By Proposition 2.5, we have that G' has depth less than $d/2$.

If $r = 1$, we are done, as we already showed that the depth of G can be reduced to less than $d/2^1$ by removing at most $(1/k) \cdot m$ edges. This covers the base case $k = 1$.

Otherwise, $r \geq 2$. Note that G' is a DAG with $m - t$ edges and depth less than $d/2 = 2^{k-1}$. Also, we have that $1 \leq r - 1 \leq k - 1$. By the inductive hypothesis ($\star$), the depth of G' can be further reduced to less than $(d/2)/2^{r-1}$ by removing at most $((r-1)/(k-1)) \cdot (m-t)$ extra edges. Thus, the depth of G can be reduced to less than $d/2^r$ by removing at most

$$\begin{aligned} t + \left(\frac{r-1}{k-1}\right) \cdot (m-t) &= \left(\frac{r-1}{k-1}\right) \cdot m + \left(1 - \frac{r-1}{k-1}\right) \cdot t \\ &\leq \left(\frac{r-1}{k-1}\right) \cdot m + \left(1 - \frac{r-1}{k-1}\right) \cdot \left(\frac{m}{k}\right) = \left(\frac{r}{k}\right) \cdot m \end{aligned}$$

edges, as required. $\square$

3 Main analysis

We are now ready to apply the aforementioned graph-theoretical results to prove Theorem 1.1, which states that a function that can be computed by a circuit of linear size and logarithmic depth can also be computed by a circuit of subexponential size and depth 3 with some interesting properties. First, we prove the following intermediate result:

Proposition 3.1. *Consider a circuit of size $O(n)$ and depth $O(\log n)$.¹ Then, for any $\epsilon > 0$, $O(n/\log \log n)$ wires can be cut to obtain a collection of subcircuits, each of which accepts $O(n^\epsilon)$ input bits.*

Proof. The circuit can be represented by a DAG G of size $c_1 n$ and depth less than $d = 2^k$ where $k = \lceil \log(c_2 \log n + 1) \rceil$ for some constants c_1 and $c_2 \geq \epsilon$. Let m be the number of edges in G . Clearly, $m \leq 2c_1 n$ since the AND/OR gates have at most 2 fan-ins. Let $r = \lceil \log(2c_2/\epsilon) \rceil$. Note that $r \geq \lceil \log(2\epsilon/\epsilon) \rceil = 1$ and $r \ll k$. By Lemma 1, we can reduce the depth of G to less than

$$\frac{d}{2^r} = \frac{2^{\lceil \log(c_2 \log n + 1) \rceil}}{2^{\lceil \log(2c_2/\epsilon) \rceil}} < \epsilon \log n + \frac{\epsilon}{c_2}$$

by deleting at most

$$\left(\frac{r}{k}\right) \cdot m = \frac{rm}{k} \leq \frac{\lceil \log(2c_2/\epsilon) \rceil \cdot 2c_1 n}{\lceil \log(c_2 \log n + 1) \rceil} = O\left(\frac{n}{\log \log n}\right)$$

edges. For each cut-wire source and the original output bit, we can construct a subcircuit by treating it as the output bit and its ancestors as input bits or gates. Clearly, each subcircuit has depth less than $\epsilon \log n + \epsilon/c_2$, and thus depends on at most $2^{\epsilon \log n + \epsilon/c_2} = O(n^\epsilon)$ input bits. $\square$

Note that the subcircuits may overlap. We are now ready to prove Valiant's depth reduction result.

Proof of Theorem 1.1. Let $\epsilon = \min\{\epsilon, 1/2\}$. By Proposition 3.1, $O(n/\log \log n)$ wires can be cut to obtain a collection of subcircuits, each of which accepts $O(n^\epsilon)$ input bits.

Fix a possible Boolean assignment to the cut wires and the output bit as 1 (in a sense, the output bit can be thought of as a 'virtual' cut wire). Then, each consistency condition is a Boolean predicate on the $O(n^\epsilon)$ original input bits (new input bits from the cut wires are fixed, so they do not count). Any such predicate has an equivalent CNF with at most $2^{O(n^\epsilon)}$ clauses, each of which excludes a falsifying assignment. We need the assignment to be consistent with the input bits. Thus, the CNFs for the $O(n/\log \log n)$ cut wires and the output bit can be ANDed to form a larger CNF of $(O(n/\log \log n) + 1) \cdot 2^{O(n^\epsilon)} = 2^{O(n^\epsilon)}$ clauses. Finally, we only need one possible Boolean assignment to work, so the larger CNFs for the $2^{O(n/\log \log n)}$ possible Boolean assignments can be OR'd to form the desired circuit of size $2^{O(n/\log \log n)} \cdot 2^{O(n^\epsilon)} = 2^{O(n/\log \log n)}$ for f .

This circuit is an OR of $2^{O(n/\log \log n)}$ CNFs, each of which has size $2^{O(n^\epsilon)} = 2^{O(n^\epsilon)}$, thus satisfying Property 1. Any given input is consistent with exactly one cut wire assignment. Thus, if the input is in the language, then exactly one of the larger CNFs outputs true. Otherwise, none of the larger CNFs outputs true. We have thus demonstrated property 2. $\square$

Remark 3.2. Interestingly, the proof shows that, after nondeterministically guessing $O(n/\log \log n)$ global bits, checking the consistency of each bit only requires reading $O(n^\epsilon)$ input bits.

This concludes our analysis of Valiant's depth reduction.

¹Per the convention, logarithms are assumed to be base 2 in these study notes.

4 Discussions

We continue by discussing the theorem’s impact and its applications beyond simply proving circuit lower bounds.

4.1 Circuit complexity

While Valiant’s result is merely a lower-bound reduction, not a finished lower-bound theorem, it has nonetheless initiated a vast line of research on the study of depth-3 circuits, which cite Valiant’s work as the main motivation. Valiant’s result has also been used as an ingredient to prove other notable circuit-complexity results [1]. That said, despite its enormous impact in the field, there is yet to be a superlinear general Boolean circuit lower bound. Valiant’s results have also spawned alternative interpretations. For example, Viola [9] observed that, for a many-output function, the cut-wire values can be viewed as a small amount of preprocessed common information. This gives rise to static data structures with small query time, whose lower bounds would imply circuit lower bounds. Golovnev, Kulikov, and Williams [3] extended Valiant’s program to show that improvements to several combinatorial objects would yield better general circuit lower bounds.

4.2 Matrix rigidity

The theorem also had an outsized impact on the topic of matrix rigidity. Suppose that a linear map $x \mapsto Ax$ is computed by a linear-size, logarithmic-depth, bounded-fan-in linear circuit over a field. After applying Valiant’s depth reduction, each output depends on the $O(n/\log \log n)$ cut wire values and surviving original inputs. Treating the cut wire values as independent inputs, the matrix A decomposes roughly as $A = L + S$ where $\text{rank}(L) = O(n/\log \log n)$ and S has $O(n^\epsilon)$ non-zeros in each row, and thus $\text{nnz}(S) = O(n^{1+\epsilon})$. If one could exhibit a matrix A that remains high-rank after changing $O(n^{1+\epsilon})$ entries, one gets a lower bound against linear-size log-depth linear circuits.

4.3 Arithmetic circuits

There are also analogous results in the study of arithmetic circuits where general arithmetic circuits can be reduced to depth-3 or -4 arithmetic circuits [2, 7, 4]. That said, the underlying mechanism is quite different in that Valiant’s depth reduction uses graph-theoretical results with nondeterminism, whereas arithmetic depth reduction exploits the structure of arithmetic operations.

5 Conclusions

In these study notes, we examined the main result of the 1977 paper titled “Graph-theoretic arguments in low-level complexity”, where Valiant showed that a linear-size logarithmic-depth Boolean circuit can be converted into a subexponential-size depth-3 circuit [8]. Valiant’s work is significant because he proposes a problem related to finding an explicit function requiring a superlinear circuit lower bound and reduces it to a problem that is seemingly easier and more accessible. We proved Valiant’s depth reduction result using his graph-theoretical proof ingredients. After our analysis, we discussed the theorem’s impact and its applications beyond simply proving circuit lower bounds. Valiant’s depth reduction leaves us with a striking challenge: Prove that shallow circuits need exponential size to show that deeper circuits need more than linear size.

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